UNIVERSITY NAME

Probability and Statistics — Final Exam

Year: 2024

Exercise 1:

In a factory, two machines M_1 and M_2 are used together to produce parts. Over a given period, the probabilities of breaking down are 0.01 and 0.008, respectively. Moreover, the probability of the event "Machine M_2 is broken down given that M_1 is broken down" is equal to 0.4.

1. What is the probability that both machines break down at the same time?
2. What is the probability that at least one machine is working?

Answer Area

Exercise 2:

Let (X, Y) be a random pair with probability density function

$$f(x,y) = \begin{cases} axy^2 & \text{if } 0 \le x \le y \le 1, \\ 0 & \text{otherwise,} \end{cases}$$

1. Calculate a.
2. Calculate the marginal densities of X and Y.
3. Give the cumulative distribution function of Y.
4. Compute the expected value of Y.
5. Calculate $P\left(X > \frac{1}{2}Y\right)$.

Answer Area

Exercise 3:

A company wants to hire an executive. n candidates apply for the position. Each of them takes a test in turn, and the first one who passes the test is hired. Let p be the probability that a candidate passes the test. Let X be the random variable defined by:

$$X = \begin{cases} k & \text{if the } k^{\text{th}} \text{ candidate is hired,} \\ n+1 & \text{if no candidate is hired,} \end{cases}$$

1. Give the probability distribution of X.
2. Calculate $E(X)$.
3. Find the minimum value of p such that there is more than a 50% chance of hiring one of the candidates.

Answer Area